StaffSync — Software Requirements Specification (SRS) & System Architecture Diagrams

Modern Human Resource Management Solution (MERN Stack: MongoDB, Express.js, React.js, Node.js) • IEEE-830 Compliant Specification, DFD Levels 0–2, UML Class Diagram & Sequence Architecture

1. Introduction & Project Overview

StaffSync — Modern Human Resource Management Solution is a full-stack enterprise web application engineered using the MERN stack (MongoDB, Express.js, React.js, and Node.js) to automate and centralize organizational HR workflows. The platform enforces strict Role-Based Access Control (RBAC) across three distinct user tiers—**HR Administrators**, **Department Managers**, and **Employees**—ensuring operational transparency, real-time attendance tracking, multi-stage leave approvals, and continuous skill development.

Project Attribute	Specification Detail
Application Name	StaffSync — Modern Human Resource Management Solution
Architecture Pattern	Three-Tier RESTful MVC + Redux Toolkit State Management (MERN Stack)
Delivered UI Variants	Variant 1: *StaffSync Aura* (Indigo Sidebar SaaS UI) & Variant 2: *StaffSync Nexus* (Emerald Dark Glassmorphic Bento UI)
Security Standard	JSON Web Token (JWT) Bearer Authentication, `bcryptjs` Salted Hashing, RBAC Middleware, & XSS Sanitization
Project Scale	6 Core Epics • 19 Functional Stories • 8 Indexed MongoDB Collections • 10 Lazy-Loaded React Views

2. Hardware, Software & Environment Requirements

To ensure reliable development, automated verification, and production deployment across Windows 10/11, macOS, and Linux environments, StaffSync adheres to the hardware and software baselines below.

Category	Component / Dependency	Minimum / Target Specification & Role
Hardware Baseline	Processor, Memory, Storage, Display	Intel Core i5 (8th Gen+) / AMD Ryzen 5, 8 GB RAM (16 GB recommended), 1 GB free disk space, 1366×768+ resolution
Backend Stack	Node.js (v16+), Express.js, Mongoose	RESTful API routing, schema validation, compound indexing, and performance telemetry middleware
Security Packages	`jsonwebtoken`, `bcryptjs`, `cors`, `dotenv`	Stateless JWT session signing (7-day expiry), 10-round salt password hashing, and cross-origin protection
Frontend Stack	React.js 18, Redux Toolkit, React Router v6, Axios	Single-Page Application (SPA) with centralized slices (`authSlice`, `hrmSlice`), lazy route splitting, and Bootstrap 5 / MUI

3. Data Flow Architecture — DFD Level 0 & Level 1

The Data Flow Diagrams (DFD) illustrate how external actors interact with StaffSync processes and persistent MongoDB data stores. **DFD Level 0** models the context boundary between Employees, Managers, HR Administrators, and the core system, while **DFD Level 1** decomposes the application into six functional sub-engines.

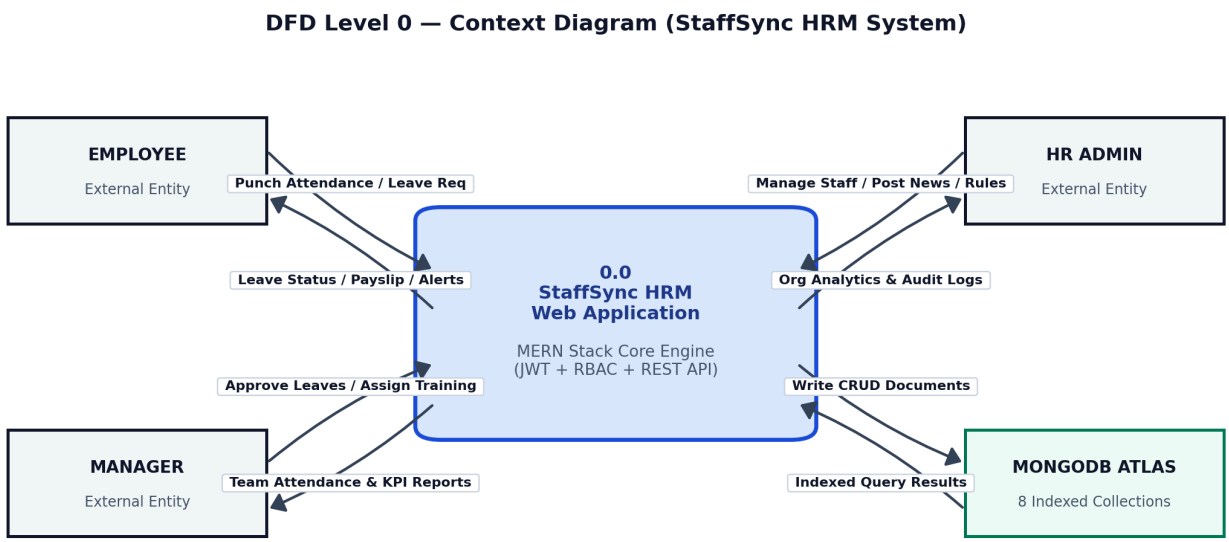


Figure 1: DFD Level 0 (Context Diagram) — External entity data exchanges between Employee, Manager, HR Admin, and MongoDB.

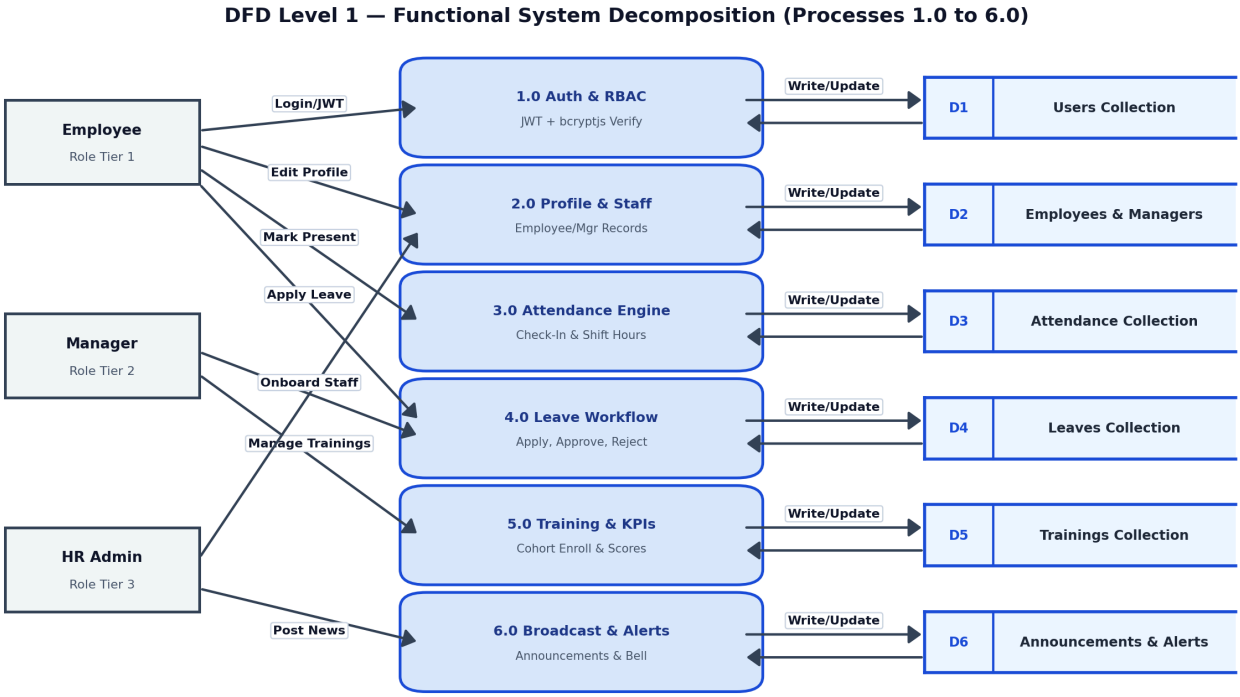


Figure 2: DFD Level 1 — Decomposition into 6 core processes (1.0 Auth to 6.0 Broadcasts) mapped to MongoDB Data Stores D1–D6.

4. Detailed Process Flow (DFD Level 2) & Functional Modules

DFD Level 2 expands Process 4.0 (Leave & Attendance Management) into four atomic validation, persistence, review, and notification steps. Below the diagram, the complete matrix maps all 19 functional tasks across the three user roles.

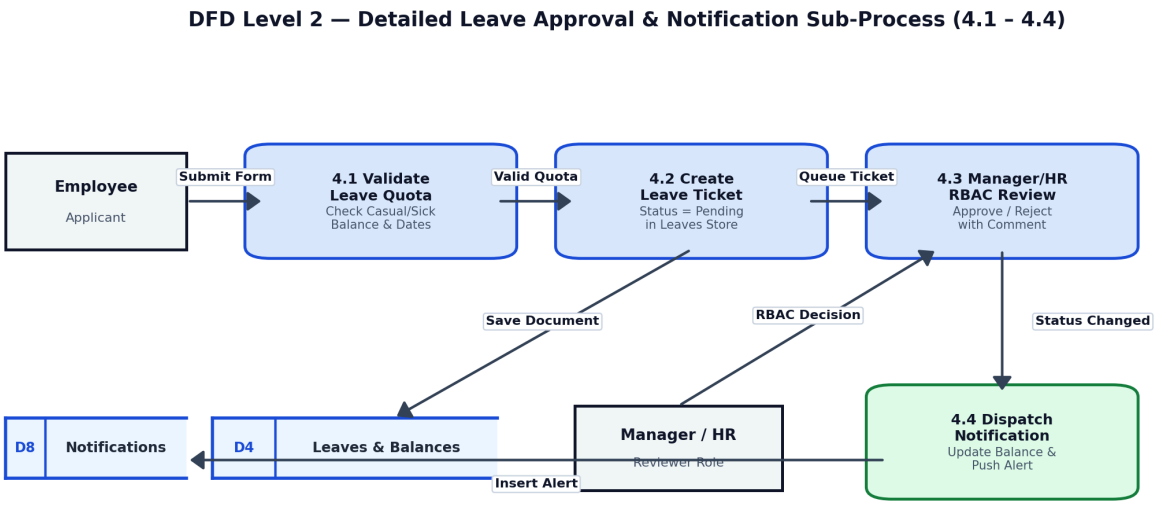


Figure 3: DFD Level 2 — Granular Leave Quota Validation (4.1), Ticket Creation (4.2), Manager/HR RBAC Review (4.3), and Alert Dispatch (4.4).

Role-Based Functional Requirements Matrix (FR-01 to FR-19)

Role Module	Req ID	Core Functional Capabilities & REST API Endpoints
Employee Module	FR-01 – FR-07	Register & JWT Login (POST `/api/auth/login`), Manage & Update Personal Profile (PUT `/api/auth/profile`), Punch Daily Attendance (POST `/api/hrm/attendance`), Apply for Leave & Track Kanban Status (POST `/api/hrm/leaves`), View Pinned Announcements, and Enroll in Training Cohorts (POST `/api/hrm/trainings/:id/enroll`).
Manager Module	FR-08 – FR-13	Supervise Department Team Members (GET `/api/hrm/employees`), Review Pending Leave Queue, Approve/Reject Leaves with Reviewer Comments (PATCH `/api/hrm/leaves/:id/status`), Launch Team Training Programs (POST `/api/hrm/trainings`), Monitor Employee Performance KPIs, and Audit Team Attendance.
HR Admin Module	FR-14 – FR-19	Onboard & Manage Employees, Oversee Department Managers (GET `/api/hrm/managers`), Audit Organization-Wide Attendance, Handle Escalated Leave Policies, Publish Official Announcements (POST `/api/hrm/announcements`), and View Executive Analytics & Payroll Estimates (GET `/api/hrm/analytics`).

5. Database Schema Dictionary & UML Class Diagram

StaffSync persists data across **8 MongoDB collections** modeled with Mongoose schemas. Compound indexes are applied on high-selectivity fields to guarantee sub-10ms query execution under pagination.

Collection	Mongoose Model	Key Fields & Foreign Key References	Optimization Indexes
Users	`User.js`	`name`, `email`, `password` (bcrypt), `role`, `department`, `status`	`{ email: 1 }` (unique), `{ role: 1 }`
Employees	`Employee.js`	`user` (→Users), `employeeCode`, `manager` (→Managers), `salary`, `performanceScore`, `leaveBalance`	`{ employeeCode: 1 }`, `{ department: 1, performanceScore: -1 }`
Managers	`Manager.js`	`user` (→Users), `managerCode`, `department`, `teamMembers[]` (→Employees), `teamRating`	`{ managerCode: 1 }` (unique), `{ department: 1 }`
Attendance	`Attendance.js`	`user` (→Users), `date` (YYYY-MM-DD), `checkIn`, `checkOut`, `workHours`, `status`	`{ user: 1, date: -1 }` (unique compound)

Collection	Mongoose Model	Key Fields & Foreign Key References	Optimization Indexes
Leaves	`Leave.js`	`user` (→Users), `leaveType`, `startDate`, `endDate`, `days`, `status`, `reviewedBy` (→Users)	`{ status: 1, createdAt: -1 }`
Announcements & Trainings & Alerts	`Announcement.js`, `Training.js`, `Notification.js`	`title`, `category`, `pinned`, `instructor`, `enrolledEmployees[]`, `recipient`, `isRead`	`{ pinned: -1, createdAt: -1 }`, `{ recipient: 1, isRead: 1 }`

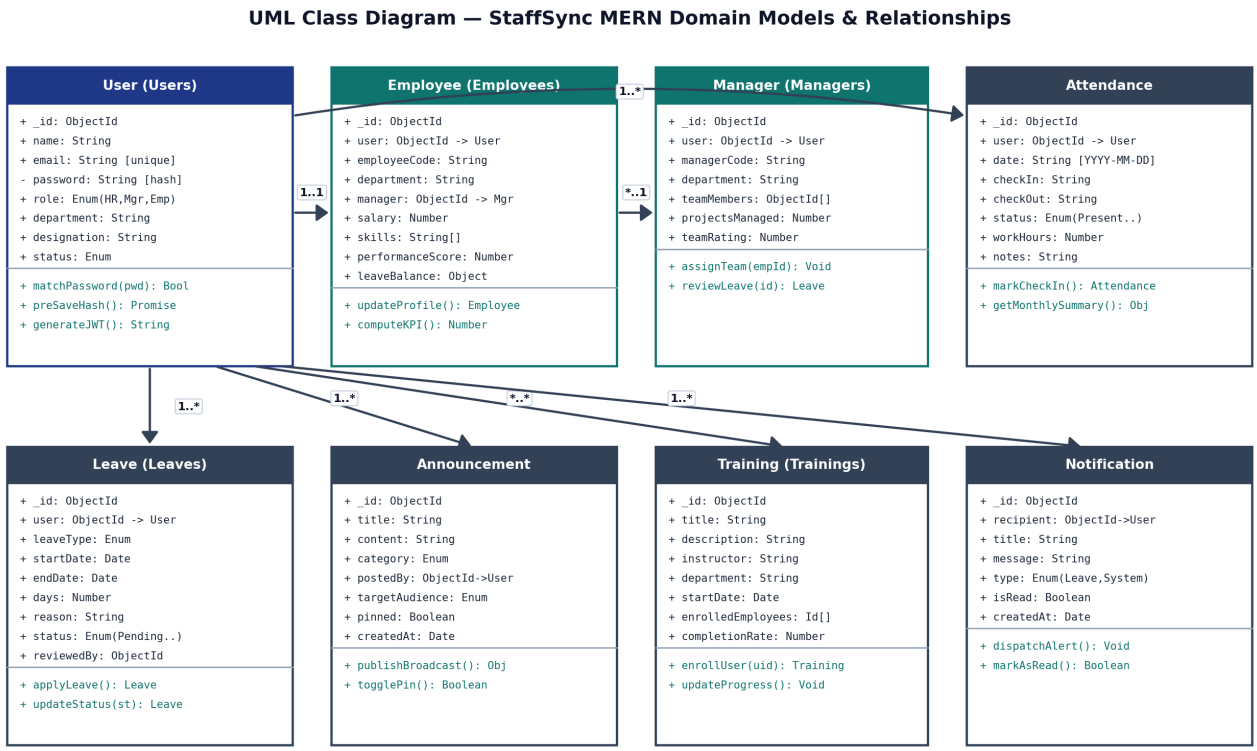


Figure 4: UML Class Diagram — Attributes, data types, methods, and multiplicity associations across all 8 MongoDB collections.

6. Runtime Sequence Architecture, Verification & UI Variants

The sequence diagram below traces an end-to-end authenticated transaction from React/Redux client login through Express JWT middleware, MongoDB persistence, manager leave approval, and real-time notification creation.

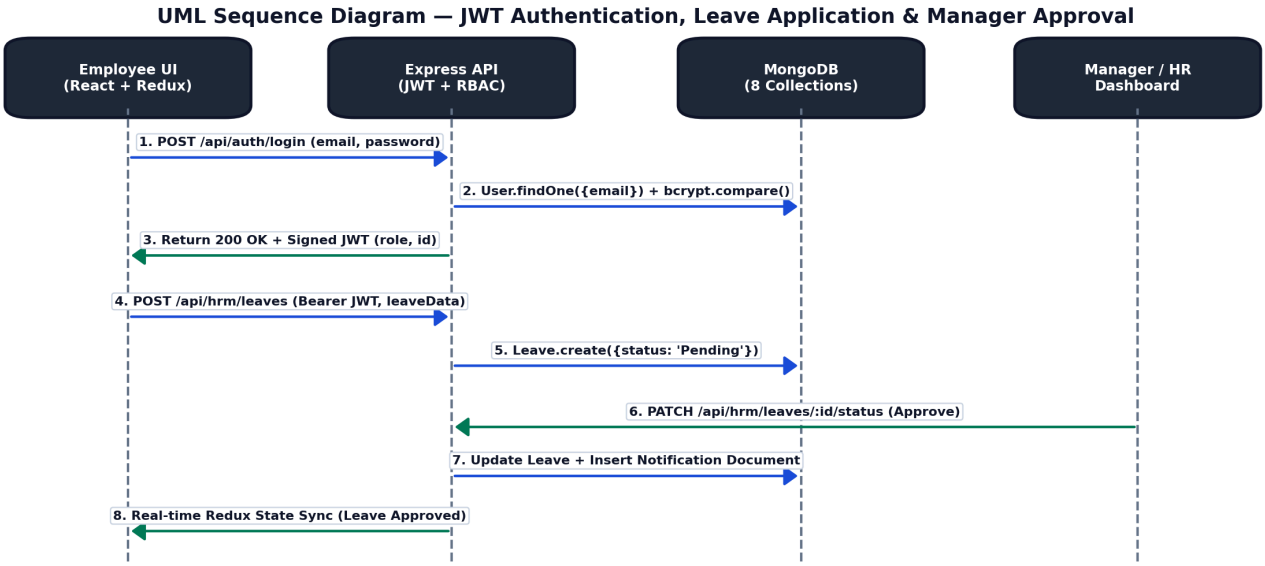


Figure 5: UML Sequence Diagram — End-to-end lifecycle of JWT Authentication, Leave Submission, and Manager/HR Approval.

Automated Verification & Dual UI Redesign Summary

Verification Suite / Deliverable	Scope Checked	Verification Outcome
1. Authentication & RBAC Suite	User schema validation, `bcryptjs` password hashing, JWT payload signing, and `authorizeRoles` middleware	✓ Verified & Passed (0 Errors)
2. Attendance & Leave Workflow Suite	Unique daily attendance punch-in, work-hours calculation, leave status transitions (Pending → Approved/Rejected)	✓ Verified & Passed (0 Errors)
3. Employee, Manager & Dashboard Suite	Employee/Manager directory creation, pinned announcements, training enrollment, and KPI aggregation	✓ Verified & Passed (0 Errors)
4. Dual UI Implementation Check	Variant 1 (*Aura Indigo Sidebar UI*) & Variant 2 (*Nexus Emerald Dark Glass UI*) live browser rendering & role switching	✓ Verified & Recorded in Demo Video